WORKSHEET

Chapter 3: Optimisation – Mixed problems

Of all the sad words of tongue or pen, the saddest are these …

8	6	10	8	7	2	6	2	5	4	9	3	9	9	1	

John Greenleaf Whittier – poet

Questions

1 Find two numbers that add to 50 and whose product is a maximum.

2 A rectangular piece of cardboard, 20 cm by 16 cm, is to be turned into an open box of maximum volume by removing a square of equal size from each corner and then folding up the sides. Determine the side length of the squares to be removed and hence find the maximum volume.

3 A company makes x computers each day at a cost of:

$$C(x) = x^3 - 10x^2 - 800x + 15\,000.$$

How many computers should be made each day to minimise costs?

4 A farmer's dog keeps straying at night, so the farmer decides to build the largest rectangular enclosure he can with 24 m of fencing. Determine the side lengths of the enclosure and hence find the maximum area enclosed.

5 Find two numbers whose difference is 14 and whose product is a minimum.

6 A can of drink is to contain 375 mL of lemonade, which is equivalent to 375 cm^3. Find the height and radius of the can (to 1 decimal place) needed to minimise the surface area and hence find the minimum surface area.

7 A particular brand of television set is sold for \$500 each in stores. The cost of manufacturing x televisions is found by:

$$C(x) = \frac{x^3}{3} + 10x^2 - 1000x + 33\,000.$$

How many sets should be manufactured to maximise profits?

8 A square-based rectangular prism has a total surface area of 54 cm^2. Determine the side lengths if the volume is a maximum and hence find that volume.

9 A bakery produces large cakes, which it sells for \$20 each. The cost of producing x cakes is found by:

$$C(x) = \frac{x^3}{3} - 20x^2 - 480x + 34\,000.$$

How many cakes should be made to maximise profits?

10 A student wants to plant a vegetable garden and needs an area of 25 m^2 to grow everything that is needed. If the garden is rectangular, determine the side lengths that would make the perimeter length a minimum and hence find that perimeter.

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Solutions

A 7 and -7

B 20

E 50

G 30

H $384\ \text{cm}^3$

I $27\ \text{cm}^3$

M $20\ \text{m}$

N 25 and 25

T $286.7\ \text{cm}^2$

V $36\ \text{m}^2$

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